

Quantitative In Situ Enhanced Oil Recovery Monitoring Using Nuclear Magnetic Resonance

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Abstract Quantitative in situ monitoring of oil recovery from sedimentary rock is demonstrated for the first time using advanced two-dimensional (2D) nuclear magnetic resonance (NMR) correlation measurements on a low field spectrometer. The laboratory-scale NMR system was chosen to provide a common physics of measurement with NMR well-logging tools. The NMR protocols are used to monitor recovery of a heavy Middle East crude oil from high permeability sandstone plugs using a brine (water) flood followed by chemical enhanced oil recovery agents: polymer and alkaline–surfactant–polymer solutions. 2D correlations between relaxation time (T_1 , T_2) and apparent self-diffusion coefficient D_{app} are used to obtain simultaneously a volumetric determination of the oil and aqueous fluid-phase saturations present in the porous material. The $T_1 - T_2$ and $D_{app} - T_2$ correlations are bulk measurements of the entire rock core-plug; excellent agreement is shown between the measures of remaining oil (from NMR) and recovered oil (from gravimetric assay of the effluent). Furthermore, we introduce the capability to measure spatially resolved T_2 distributions on a low field spectrometer using a rapid frequency-encoded $\gamma - T_2$ map. A non-uniform distribution of remaining oil is observed due to viscous instabilities in the flowing liquids; the final oil saturation ranges from $S_o^{(final)} \approx 0$ to 20 % along the direction of flow. These results highlight the quantitative nature of the NMR data obtainable in low field NMR core analysis and also the importance of spatially resolved measurements when studying short core-plugs.

Keywords Nuclear magnetic resonance · Core analysis · Enhanced oil recovery · Spatially resolved T_2 · Diffusion-editing

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1 Introduction

Nuclear magnetic resonance (NMR) is a popular, non-invasive technique for probing fluid transport in porous media (Watson and Chang 1997). A recent review (Gladden and Mitchell 2011) highlights the importance of NMR measurements including imaging, flow propagators, and relaxation time analysis in multi-phase systems such as heterogeneous catalysts, fluidized beds, and sedimentary rocks. One of the key advantages of NMR over other methods of probing imbibed fluid behavior is the potential for quantitative observations. In the context of this work, we consider “quantitative” to mean good agreement between fluid saturations determined via NMR (and hence total porosity) and a conventional estimate of the same, e.g., a gravimetric determination of porosity and saturation state.

Quantitative determination of fluid saturation is particularly important in the petroleum industry when estimating oil and gas reserves in reservoirs. NMR is an important well-logging measurement due to its ability to probe the fluid content of the rock matrix in close proximity to the well-bore, thereby providing information on porosity ϕ and permeability κ of the rock (Kenyon 1997). To obtain an accurate interpretation of NMR well logs, it is often desirable to complement the field data with laboratory-scale core analysis. Recently we supported a single-well in situ enhanced oil recovery (EOR) field trial of chemical EOR agents with NMR data obtained in the laboratory (Arora et al. 2010).

NMR fluid characterization in small plugs of cored reservoir material is conducted typically on low field strength permanent magnets (i.e., a field strength $B_0 \ll 1$ T) to provide comparability to well-logging data. The main limitation on these measurements is the sample volume. Capillary end effects and geometric constraints (e.g., inlet and outlet geometry) can be significant on the plug length-scales that can be accommodated practicably in a commercial NMR magnet. It is therefore important to acknowledge both the advantages and limitations of these laboratory-scale core analyses. One-dimensional (1D) longitudinal T_1 relaxation time distributions have been shown previously to provide quantitative water (brine) and oil (decane) saturations in chalk plugs (Howard 1995). Here we demonstrate for the first time how quantitative estimates of oil and aqueous fluid-phases can be obtained robustly from more sophisticated experiments: namely two-dimensional (2D) correlations of longitudinal T_1 and transverse T_2 relaxation time (Song et al. 2002), and apparent self-diffusion coefficient D_{app} and T_2 relaxation time (Hürlimann et al. 2002). D_{app} is reduced from the free liquid diffusion coefficient D_0 due to restrictions imposed on the mean diffusion path length by the porous matrix.

In this work, we present an EOR study to demonstrate the quantitative nature of a range of NMR techniques. The focus of this article is, therefore, the application of these NMR techniques at very low magnetic field strengths within core analysis, rather than the liquid transport and oil recovery mechanisms per se. In future work, we will consider details of the EOR process monitored using the NMR methods validated here.

A heavy Middle East crude oil was chosen to provide good relaxation and diffusion contrast with the brine and EOR chemicals. As a proxy for the poorly consolidated material typical of the reservoir, we selected Bentheimer sandstone. We have used this outcrop rock previously in studies of magnetic susceptibility induced “internal gradients” (Mitchell et al. 2010) and flow propagators (Mitchell et al. 2008a, b). Polymer and alkaline-surfactant-polymer (ASP) solutions were injected to enhance the oil recovery following the initial brine flood.

The determination of saturation via NMR can be achieved readily in bulk measurements on the length-scale of a core-plug. In early work, the quantitative conversion from NMR signal to oil saturation was demonstrated by Saraf and Fatt (1967) who used D_2O to eliminate the ^1H

signal from the aqueous fluid-phase, enabling the unambiguous detection of oil remaining in a Boise sandstone plug. However, there is a general paucity of published data confirming the expected linear response of NMR signal amplitude with liquid volume over a wide range of saturation states in core-plugs. Furthermore, to provide a good understanding of NMR measurements of remaining oil it is necessary to consider the spatial distribution of the liquid. A critical limitation in NMR imaging of porous materials is the relaxation weighting inherent in most standard pulse sequences (Haacke et al. 1999) as, typically, a 1D profile (or a single dimension in a multi-dimensional image) is acquired with a frequency-encoded spin echo (Callaghan 1991). Frequency encoding has the advantage of providing a profile in a single scan where the resonant frequency of the spins is spatially encoded via the application of a magnetic field gradient of constant amplitude. However, each k -space datum (where k -space is the Fourier reciprocal of real space) is obtained at a different time; therefore, each k -space point will have a different relaxation time weighting. This relaxation time weighting can be overcome by (a) recording a zero-gradient echo shape to deconvolve from the final profile and (b) combining the frequency-encoded spin echo profile with a Carr–Purcell Meiboom–Gill (CPMG) echo train (Carr and Purcell 1954; Meiboom and Gill 1958). Accordingly, we employ a frequency-encoded profiling measurement to resolve the T_2 distribution as a function of plug length oriented along the y -axis. These $y - T_2$ maps show the variation in final oil saturation $S_o^{(\text{final})}$ and reveal the extent of end effects in the laboratory-scale measurement.

An alternative method for measuring $y - T_2$ maps is based on phase-encoded single-point imaging (SPI) protocols (Li et al. 2009; Petrov et al. 2011). However, a spin echo SPI $y - T_2$ map requires each k -space point to be sampled in a separate measurement. As such, this acquisition method is slower than the frequency-encoded spin echo $y - T_2$ technique employed here. Although acquisition time was not a critical factor in these studies, we wished to validate the frequency-encoded method for future work in dynamic systems.

Since the 1980s, magnetic resonance imaging (MRI) (Haacke et al. 1999) has been a popular method for obtaining a localized volume average of liquid saturation in rock core-plugs, see for example (Brown 1988; Mahmoud et al. 1990; Chardaire-Rivière and Roussel 1991; Mattiello et al. 1997; Baldwin and Spinler 1998; Chen et al. 2003; Fernø et al. 2007) plus other references given in this article. Fluid-phase discrimination between the protonated oil and aqueous fluid-phases has been achieved in several ways, including relaxation time weighting (Blackband et al. 1986), paramagnetic doping (Baldwin and Yamanashi 1989), or chemical shift selectivity (Edelstein et al. 1988; Enwere and Archer 1992; Borgia et al. 1994; Nørgaard et al. 1995; Doughty and Tomutsa 1996). Another approach is the suppression of the aqueous fluid-phase signal using D_2O (Brautaset et al. 2008; Green et al. 2008), as demonstrated in earlier work (Saraf and Fatt 1967). Attempts have also been made to image pore-scale structures within the rock indirectly (Gleeson et al. 1993), although due to the inherent limited spatial resolution of MRI in heterogeneous materials (typically $\Delta y \sim 100 \mu\text{m}$, Mitchell et al. 2006), it is not possible to visualize the content of individual pores. Continued investment in NMR technique development over the last decade means MRI methods are instead being applied to the determination of petrophysical properties such as capillary pressure curves (Baldwin and Yamanashi 1991a; Borgia et al. 1994; Nørgaard et al. 1995; Chen and Balcom 2005; Green et al. 2008). Accordingly, MRI is now of renewed interest in core analysis.

The majority of early MRI studies of rocks were implemented on intermediate to high field strength magnets ($B_0 > 1 \text{ T}$) based on medical imaging systems (Edelstein et al. 1988; Doughty and Maerefat 1989; Fordham et al. 1993; Gleeson et al. 1993; Borgia et al. 1994; Fordham et al. 1995; Nørgaard et al. 1995), though some experiments were conducted on magnets with field strengths as low as $B_0 = 0.5 \text{ T}$ (Blackband et al. 1986; Baldwin and

Yamanashi 1989, 1991a, b). On such systems, spatial resolution on the order of millimetres is obtainable. In core analysis, very low magnetic field strengths are preferred ($B_0 \sim 50$ mT) to provide comparability to well-logging tools. In addition, the use of low field magnets reduces the influence of magnetic susceptibility induced internal gradients on measurements of T_2 (Hürlimann 1998); the effective internal gradients, which scale with B_0 , are known to perturb the diffusion-sensitive T_2 measurement (Mitchell et al. 2010) and prevent quantitative saturations being obtained from MRI data. The inherent poor signal-to-noise ratio (SNR) associated with low field magnets ensures the resolution will be limited to a few millimeters at best. This is not a serious concern when weighed against the advantages of a quantitative signal intensity that can be rescaled to saturation and porosity. Even on high field magnets it is not possible to obtain pore-scale resolution (as noted above), and so the absolute resolution becomes unimportant. Here, we focus on obtaining fluid-phase discrimination through relaxation time contrast. Relaxation mapping on intermediate field strength magnets has relied primarily on T_1 contrast (Fordham et al. 1993; Borgia et al. 1994; Fordham et al. 1995; Borgia et al. 1999; Bortolotti et al. 2009) which is unaffected by the presence of strong internal gradients (Washburn et al. 2008). However, T_2 is the metric of choice in petrophysics as this is the basis for NMR well-logs. As such, it is important in laboratory-scale petrophysical applications to concentrate on T_2 mapping methods that provide quantitative data on low field spectrometers.

The NMR methodologies and results presented here are differentiated from the existing literature through a combination of key factors:

1. All the NMR measurements are performed on a low field spectrometer ($B_0 = 42$ mT) so as to be comparable with well-logging tools.
2. Quantitative data is obtained from a range of acquisition protocols, including 1D T_2 measurements, and 2D $T_1 - T_2$ and $D_{\text{app}} - T_2$ correlations. We use these data to obtain independent measurements of saturation and compare these results to conventional gravimetric estimates of remaining oil. Our results confirm the linear relationship between NMR signal amplitude and liquid volume expected over a wide range of saturation states.
3. Relaxation and diffusion contrast is exploited to measure simultaneously the oil and aqueous fluid-phase saturations.
4. Spatially resolved saturation profiles are obtained on a low field spectrometer using frequency-encoded $y - T_2$ maps.

2 Materials

The core-plugs used in this study were Bentheimer sandstone: a well-consolidated German outcrop stone used commercially in the construction industry. This rock was selected in lieu of actual reservoir material which was a poorly consolidated sand. Bentheimer is an aeolian sandstone, massive with large grains, uniform texture, and high permeability, that retained its structure over the course of the experiments. The cylindrical plugs had dimensions of $60 \text{ mm} \times 38 \text{ mm}$ (length $\times$ diameter) with a porosity of $\phi \approx 23\%$ and gas permeability of $\kappa \approx 3 \mu\text{m}^2$, determined prior to liquid saturation. Two plugs (referred to herein as “B1” and “B2”) with consistent properties were selected from the available material. These plugs were saturated initially under vacuum with a synthetic brine.

The synthetic brine, having the same chemical composition for both the formation and injection water, was prepared to be equivalent to that found in the reservoir. The low-salinity

brine had a total dissolved solid (TDS) content of $\sim 4,800$ ppm. At the typical reservoir temperature of 46.4°C the brine had a density of $\rho_w = 0.993\text{ g cm}^{-3}$.

The dead crude oil was diluted with 5.9 wt% 1,2,3,4,5,6,7,8,9,10-decahydronaphthalene (as supplied) to provide a viscosity similar to that of the live crude in the reservoir. At the reservoir temperature of 46.4°C the oil had a viscosity of $\mu_o \approx 105\text{ mPa s}$ determined at a shear rate of 7.1 s^{-1} . The density of the oil at the same temperature was $\rho_o = 0.897\text{ g cm}^{-3}$.

The brine-saturated plugs were spun under oil in a centrifuge for 20 h at $\omega = 471\text{ s}^{-1}$ (maximum capillary pressure $p_c^{(\text{max})} = 94\text{ kPa}$) to provide an initial oil saturation of $S_o^{(\text{init})} \approx 91\%$. The plugs were then aged under crude oil at 46.4°C for 2 months in an attempt to restore reservoir wettability to the sandstone surface. Despite the aging process, an independent imbibition experiment showed that the Bentheimer still imbibed water spontaneously against oil, indicating the rock surface remained preferentially water wet.

Two stages of EOR flood were used in this study, comprising a polymer-only stage and an ASP stage. Both the polymer and ASP solutions were used as provided. At 46.4°C , the density of the polymer solution was $\rho_p = 0.994\text{ g cm}^{-3}$ and the density of the ASP solution was $\rho_{\text{ASP}} = 1.004\text{ g cm}^{-3}$. The ASP solution was seen to be clear and colorless, whereas the polymer solution appeared cloudy (white tint) in a large volume. However, small quantities of the polymer solution were noted to be clear and colorless. Any particulates in the polymer solution were considered small compared to the mean pore throat size R_x of the Bentheimer sandstone, determined by mercury intrusion porosimetry (MIP) to be $R_x \approx 45\text{ }\mu\text{m}$. Therefore blockage of pore throats by polymer particulates was considered unlikely. Very low injection pressures observed throughout the recovery experiment supported this assumption. Interfacial tensions (IFT) between the liquid components were determined using a DataPhysics SVT20N spinning drop tensiometer. The oil/brine IFT $= 5.5\text{ mN m}^{-1}$ and the oil/polymer solution IFT $\approx 9.7\text{ mN m}^{-1}$.

2.1 Electrical Conductivity

Due to the configuration of the flow-through NMR compatible core-holder, electrically conductive liquids in the sample will provide a route for noise injection into the otherwise shielded radio frequency (rf) probe used for detection in the NMR spectrometer. Under normal operating conditions, this rf interference (noise) would be screened from the probe. However, a continuous conductive path from the inlet to the outlet of the core-holder acts as a rf antenna, thereby reducing the SNR of the acquired data. To assess the potential impact of the flood liquids, the electrical conductivity σ of the aqueous fluid-phases was determined. The synthetic brine had $\sigma = 7,890\text{ }\mu\text{S}$, the polymer solution exhibited $\sigma = 7680\text{ }\mu\text{S}$, and the ASP solution had $\sigma = 17,850\text{ }\mu\text{S}$. For comparison, deionized water available in the laboratory gave $\sigma = 1.4\text{ }\mu\text{S}$. All electrical conductivities were determined at a nominal temperature of 15°C . Despite attempts to ground the flow system, the SNR of the NMR data decreased as the electrical conductivity of the injected liquid increased. Based on a comparison of the real and imaginary channels in the phase-corrected T_2 data (see Sect. 3.1), the SNR decreased from $\text{SNR} \sim 150$ to $\text{SNR} \sim 60$ over the course of a flood. A discussion on noise in low-field NMR experiments is given by Singer et al. [2006]. This reduction in SNR is not a fundamental limitation in the experiment as the analysis methods used to process the NMR data (see Sect. 3) provide stable and reproducible solutions in the presence of noise.

3 Methods

The NMR system comprised a Maran DRX2-HF spectrometer and a “Big-2” permanent magnet (rock core analyzer)¹ with a field strength of $B_0 = 42$ mT (nominal resonant frequency $\nu_0 = 1.93$ MHz for ^1H). The magnet was fitted with a single y-axis field gradient coil providing a maximum gradient $g_y^{(\max)} = 0.5$ T m⁻¹ and a rf probe with a bore diameter of 53 mm. Rf pulse durations of $t_{90} = 15$ μs and $t_{180} = 30$ μs were typical for tip angles of 90° and 180°, respectively. A custom-made core-holder² was used to mount the rock sample inside the NMR magnet/probe arrangement. The core-holder provided temperature and confining pressure regulation of the sample through a per-fluorinated oil. Sample temperature, determined by fiber optic sensors, was maintained at an average 46.4°C with an isostatic confining pressure of 5 MPa throughout. Pressure at the inlet face of the plug was monitored continually with the outlet at atmospheric pressure.

The oil recovery experiments were performed in a measure-inject-measure manner, equivalent to the log-inject-log protocol of a pilot operation (Arora et al. 2010). A known volume of flood liquid (corresponding to some fraction of the total pore volume PV in the plug) was injected. The sample was then held at static conditions while the NMR data were acquired. The next volume of flood liquid was injected, and the NMR measurement repeated. Each recovery experiment comprised eight stages: initial, four brine injections of 0.16 PV each, one polymer solution injection of 0.5 PV, an ASP solution slug (0.3 PV), and two more polymer solution injections of 1 PV each. Plug B1 had a pore volume of 1 PV $\equiv 15.16$ cm³ and plug B2 had a pore volume of 1 PV $\equiv 15.30$ cm³, determined by He porosimetry. As such, the absolute volume of flood liquid differed very slightly in each experiment, but the proportions relative to the plug pore volume remained constant.

All flow lines were filled with synthetic brine initially. Excess oil was wiped carefully from the face of the plug prior to being loaded into the core-holder. Before any flooding began, the core-plug was allowed 1 h to reach thermal equilibrium. The application of the confining pressure and thermal expansion resulted in a small quantity of liquid (oil) being expelled from the core-plug. An initial flush of excess polymer solution was applied to clear all exit flow lines before any data were taken or any flood liquids were pumped. Effluent was collected from the top exit of the core-holder at atmospheric pressure. As few volatiles were present in the crude oil, back pressure and confinement of the effluent was not required. At each stage of the flood, effluent was collected in weighed centrifuge tubes. Additional effluent in the upper flow lines was recovered by polymer solution flush to improve the accuracy of the gravimetric oil assay.

The synthetic brine and polymer solutions were pumped through the core-plug at a mean volumetric flow rate of $\dot{V} = 1.0$ cm³ min⁻¹. This flow rate corresponded to a pore velocity of $u = 6.25 \times 10^{-5}$ m s⁻¹. A continuous pressure drop not exceeding $\Delta p \simeq 138$ kPa was raised across the rock. For the ASP solution slug injection, a ground-glass gas-tight syringe was used to inject the slug volume by hand, creating a maximum pressure drop of $\Delta p \simeq 550$ kPa. The dimensionless microscopic capillary number for the brine was $\text{Ca}_w = 6.88 \times 10^{-6}$ and for the polymer solution $\text{Ca}_p = 8.33 \times 10^{-4}$. These microscopic capillary numbers are converted to macroscopic capillary numbers of $\bar{\text{Ca}}_w = 0.017$ and $\bar{\text{Ca}}_p = 2.00$ using the dimensionless ratio of sample length to pore radius as described by Hilfer and Øren [1996].

At the end of the experiment, the effluent was centrifuged to provide good separation of the oil and aqueous (brine/polymer solution) fluid-phases. The aqueous fluid-phase was

¹ NMR system manufactured by Oxford Instruments, Abingdon, UK

² Core-holder manufactured by ErgoTech, Conwy, UK

then aspirated from the centrifuge tubes; the high viscosity of the oil at ambient temperature facilitated clean extraction of the lower fluid-phase. The residual water content in the oil was estimated at $\ll 1$ % w/w by Karl–Fischer titration (consistent with the crude oil as supplied). The recovered oil volumes were determined gravimetrically.

A suite of NMR measurements were performed at each stage in the recovery experiments: a 1D T_2 relaxation time distribution (with variable echo spacing), a 2D $T_1 - T_2$ correlation, a 2D $D_{\text{app}} - T_2$ correlation, and a 2D $y - T_2$ map. The NMR data were acquired in 75 min, with one entire oil recovery experiment lasting 14 h. Each of these NMR experiments is described below. A review of 2D inversion methods used to process relaxation time and diffusion data has been presented elsewhere by Mitchell et al. [2012].

The experimental procedures for acquiring T_2 , $T_1 - T_2$, and $D_{\text{app}} - T_2$ distributions are described in Sects. 3.1 through 3.3, respectively. The aim of each of these three experiments is the determination of a single parameter: the total magnetization $m(0)$ (at time zero) associated with the fluid in the core-plug. Due to the inherent relaxation time and diffusion contrast between the heavy oil and aqueous fluid-phases, we can associate a fraction of $m(0)$ with the oil, and the remainder with the aqueous fluid-phase. However, because the magnetization m is time dependent, the determination of $m(0)$ is not straight-forward. All NMR experiments have a dead-time associated with the acquisition of the first datum and, depending on the nature of the acquisition, this dead-time can encompass a significant degree of nuclear spin relaxation (i.e., signal decay) meaning the initial datum amplitude will not reflect accurately the total fluid volume, and hence saturation state, of the sample. It is therefore imperative that we fit the data in an appropriate manner, allowing an extrapolation of m back to zero time. In natural porous materials, such as reservoir rocks, it is accepted that the relaxation or diffusion decay can be modeled by a continuous sum of exponential components (Davies and Packer 1990a, b), each with an associated magnetization (amplitude) and relaxation/diffusion (decay rate) constant. Typically, the determination of a large number of exponential amplitudes and rate constants is an ill-posed problem: there are more unknown parameters than data. In NMR processing, it is now common practice to use a numerical inversion that provides a stable probability distribution of relaxation times or diffusion coefficients by assuming the distribution is smooth, non-negative, and is constrained by pre-defined limits of relaxation time or diffusion coefficient. The area under the probability distribution is proportional to the total magnetization at zero time, $m(0)$. Therefore, we obtain our oil and aqueous fluid-phase saturations simply by integrating over the region of the probability distribution associated with oil or brine/polymer solution. The exact same process is applied to both the 1D and 2D probability distributions; in 2D we integrate the “volume” under the probability distribution to obtain $m(0, 0)$: the magnetization at zero time in both dimensions. Example data sets for each NMR experiment are included in Sect. 4 to demonstrate the processing and methodology for determining the fluid saturation from these NMR experiments.

3.1 Relaxation Time Analysis: T_2 Distributions

The measurement of T_2 using the CPMG experiment is a standard feature of NMR core analysis. T_2 distributions are also measured routinely in down-hole well logging, although the pulse sequence requires some optimization for use on well-logging tools due to the grossly inhomogeneous magnetic field of the unilateral device (Hürlimann and Griffin 2000). Here, the relatively homogeneous magnetic field of the bench-top magnet enables the use of the standard CPMG sequence as described by Meiboom and Gill [1958]. A 90° rf pulse rotates the coherent spin ensemble away from its equilibrium position in the B_0 magnetic field in order to be detected. The spin ensemble then loses coherence as individual spins experience

different local magnetic fields and hence precess at slightly different angular frequencies. As only the magnitude of the spin ensemble is observed by the experimentalist, this loss of coherence appears as a reduction in signal intensity with increasing time. Ideally, the decay of magnetization should be determined only by coupling between nearby spins (proton or electron). However, spatial variations in B_0 also enhance the dephasing (signal decay) rate. The application of a 180° rf pulse at time τ_2 reverses the dephasing of the spin ensemble, so the spin ensemble refocuses to its coherent state. This is observed as a so-called spin echo (Hahn 1950). If the 180° rf pulse is applied repeatedly (each pulse separated in time by $2\tau_2$), then an echo train is formed. Assuming the translational motion (i.e., diffusion) of the spins is small compared to the spatial variations in B_0 , the CPMG echo train will remove the influence of magnetic field inhomogeneities on T_2 . The CPMG pulse sequence is also used for data acquisition in the 2D $T_1 - T_2$ and $D_{\text{app}} - T_2$ experiments.

Here, T_2 decays were acquired using the standard CPMG echo train (see Fig. 1), with an inter-echo time ranging from $t_E = 2\tau_2 = 200$ to $1,000 \mu\text{s}$. A recycle delay of 10 s was present between each acquisition. Four repeat scans were stored for signal averaging, with 11 points acquired from the top of each echo using a dwell time of $t_{\text{dw}} = 1 \mu\text{s}$ (the time over which the continuous analogue rf signal is summed to form a discrete digital data point) and averaged to improve the SNR. The variable t_E measurement provides an estimate of internal gradient strength following the theory of Hürlimann [1998]. Here, an average effective internal gradient of $g_{\text{eff}} \approx 0.03 \text{ T m}^{-1}$ was found and considered negligible for the purposes of these analyses. A T_2 relaxation time distribution is obtained from the CPMG data using a numerical inversion method. The exponential decay of the normalized magnetization m at time nt_E (where n is the number of echoes) is described by the first kind Fredholm integral equation

$$\frac{m(nt_E)}{m(0)} = \int_0^\infty \exp\left\{-\frac{nt_E}{T_2}\right\} f(\log_{10} T_2) d\log_{10} T_2 + \epsilon, \quad (1)$$

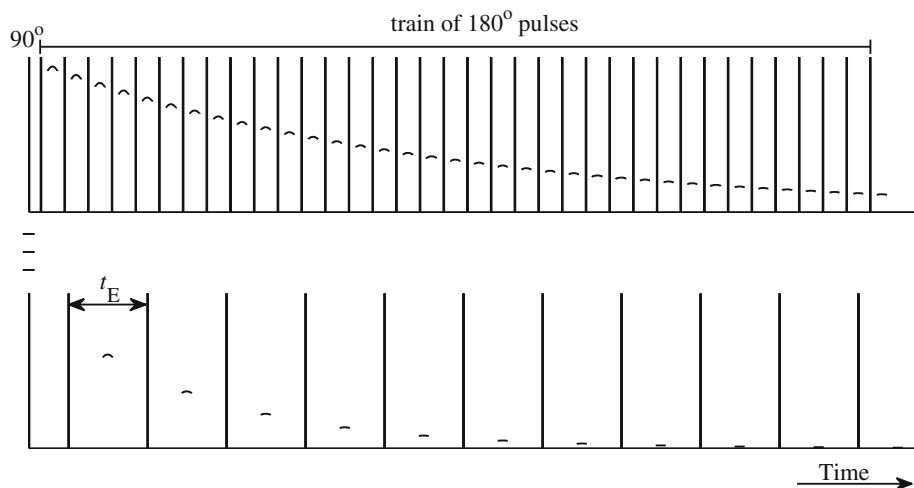


Fig. 1 Timing diagram for the variable t_E CPMG sequence with (top) short t_E and (bottom) long t_E . CPMG decays are acquired in a series of separate experiments as t_E is increased. Vertical bars represent rf pulses. The CPMG decays shown are truncated for clarity

where $f(\log_{10} T_2)$ is the required distribution, determined on the $\log_{10} T_2$ scale using Tikhonov regularization via the method given by Wilson [1992] and ϵ encompasses the experimental error (noise).

3.2 Relaxation Time Analysis: $T_1 - T_2$ Correlations

$T_1 - T_2$ correlations were determined from the saturation-recovery CPMG sequence given by Song et al. [2010], see Fig. 2. Good saturation of the spin ensemble is achieved via the application of a comb of 90° rf pulses separated by a variable time increment and a homospoil gradient pulse. Once the spin ensemble has been saturated, it is allowed to recover with T_1 for a time τ_1 , and then a standard CPMG sequence is applied to acquire n echoes with a fixed echo spacing of $t_E = 600 \mu\text{s}$; other acquisition parameters were as given in Sect. 3.1. Saturation recovery is preferable to inversion recovery in this case because (a) good saturation is easier to achieve than good inversion in the inhomogeneous B_0 and rf magnetic fields, and (b) the recycle time between successive acquisitions can be reduced to a negligible duration (here, 1 s). As long as the spin ensemble is saturated at the end of the comb of 90° rf pulses, the starting condition is irrelevant. Four repeat scans were acquired for signal averaging, with 32 different τ_1 delays ranging from $\tau_1^{(1)} = 1 \text{ ms}$ to $\tau_1^{(32)} = 8 \text{ s}$. Each 2D $T_1 - T_2$ data set was acquired in 10 min. The data are described by the first-order Fredholm integral equation

$$\frac{m(\tau_1, nt_E)}{m(0, 0)} = \int_0^\infty \int_0^\infty k_0(\tau_1, nt_E, T_1, T_2) f(\log_{10} T_1, \log_{10} T_2) d\log_{10} T_1 d\log_{10} T_2 + \epsilon, \quad (2)$$

where the kernel function k_0 is separable under $k_0 = k_1 \times k_2$ such that

$$k_1 = 1 - \exp\left\{-\frac{\tau_1}{T_1}\right\}; \quad k_2 = \exp\left\{-\frac{nt_E}{T_2}\right\}. \quad (3)$$

Accordingly, the $T_1 - T_2$ correlation can be obtained efficiently following the inversion method of Venkataramanan et al. [2002], using Tikhonov regularization as described by Butler et al. [1981] with the optimum degree of smoothing chosen by the generalized cross validation (GCV) method (Wahba 1977).

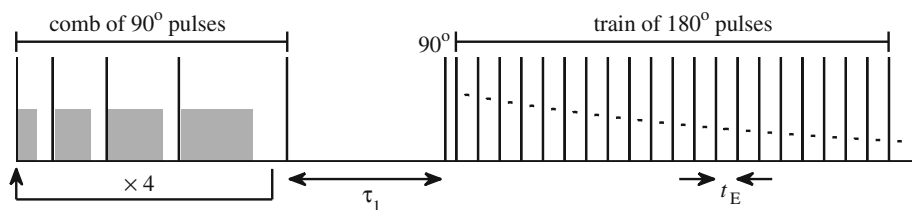


Fig. 2 Timing diagram for the saturation-recovery CPMG sequence. Vertical bars represent rf pulses. Gray rectangles represent magnetic field gradient homospoil pulses. Saturation is achieved via a comb of 90° rf pulses. The longitudinal recovery time τ_1 is increased in successive experiments. The CPMG echo train shown is truncated for clarity

3.3 Diffusion-Editing: $D_{\text{app}} - T_2$ Correlations

A range of diffusion measurements are available for core analysis, ranging in complexity from the basic pulsed field gradient (PFG) spin echo technique of [Stejskal and Tanner \[1965\]](#) to the alternating pulsed gradient stimulated echo (APGSTE) of [Cotts et al. \[1989\]](#), favored in high field NMR of porous materials due to the reduction in internal gradient effects. At low magnetic field strengths, and given the weak internal gradients present in our samples, such complicated pulse sequences are not required. Instead, we opt for the tool-equivalent PFG double spin echo (PFG-2SE) sequence described by [Mitchell and Fordham \[2011\]](#), see Fig. 3. The PFG-2SE experiment provides similar performance to the fixed field gradient (FFG) double spin echo “diffusion-editing” sequence available on well-logging tools ([Hürlimann and Venkataramanan 2002](#)). Gradient pulses of duration δ are applied around the first two refocusing pulses of a CPMG echo train. By increasing δ (and hence the initial echo spacing t_{SE}) in subsequent acquisitions, a correlation between apparent diffusion coefficient and T_2 is obtained. The CPMG portion of the echo train has the same parameters as described in Sect. 3.1. The gradient duration varied from $\delta^{(1)} = 0$ ms (equivalent to a standard CPMG acquisition, Fig. 1) to $\delta^{(32)} = 10$ ms. The gradient amplitude $g_{\text{PFG}} = 0.48 \text{ T m}^{-1}$. A recycle delay of 10 s was used with four repeat scans for signal averaging.

The processing required for the $D_{\text{app}} - T_2$ data is slightly different from that used for the $T_1 - T_2$ data. The form of the data is described by a similar first-order Fredholm integral equation

$$\frac{m(t_{\text{SE}}, nt_{\text{E}})}{m(0, 0)} = \int_0^\infty \int_0^\infty k_0(t_{\text{SE}}, nt_{\text{E}}, D_{\text{app}}, T_2) f(\log_{10} D_{\text{app}}, \log_{10} T_2) \times d \log_{10} D_{\text{app}} d \log_{10} T_2 + \epsilon, \quad (4)$$

where k_0 has the form

$$k_0 = \exp\{-bD_{\text{app}}\} \exp\left\{-\frac{2t_{\text{SE}} + nt_{\text{E}}}{T_2}\right\}, \quad (5)$$

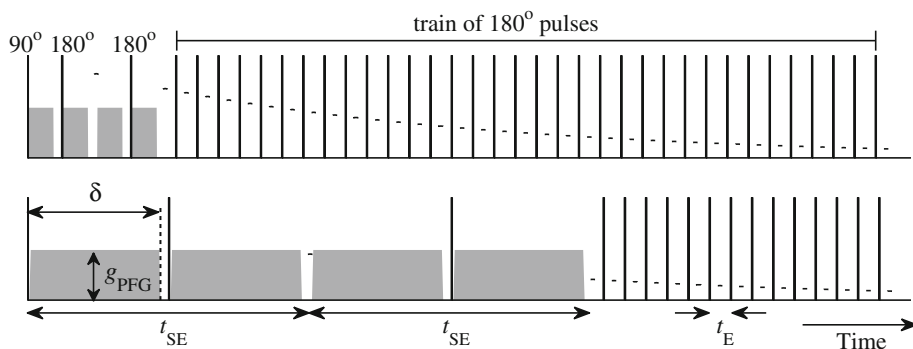


Fig. 3 Timing diagram for the PFG-2SE diffusion-editing sequence with (top) short δ and (bottom) long δ . Vertical bars represent rf pulses. Gray rectangles represent magnetic field gradient pulses. The gradient pulse amplitude is g_{PFG} . Both δ and t_{SE} are increased in successive experiments. The CPMG echo trains shown are truncated for clarity

with

$$b = 2 (\gamma_{\text{p}} g_{\text{PFG}} \delta)^2 \left(t_{\text{SE}} - \frac{\delta}{3} \right), \quad (6)$$

for the square gradient pulses shown in Fig. 3 and assuming the internal gradient strength $g_{\text{eff}} \rightarrow 0$; γ is the gyromagnetic ratio for ^1H . Performing the numerical inversion on these data is expensive computationally as k_0 is not a separable function. The size of the problem is reduced by an initial data compression stage (window averaging) as detailed by Mitchell and Fordham [2011]. Here we chose to invert the data using a maximum entropy inversion given by Heaton [2005], again to provide consistency with processing methods used in well logging. The maximum entropy inversion has been shown elsewhere to give consistent results to Tikhonov inversion, albeit with smoother (i.e., broader) peaks in the final $D_{\text{app}} - T_2$ correlation (Mitchell et al. 2012).

3.4 Spatial Resolution: $y - T_2$ Maps

Spatial position y is encoded in the reciprocal (Fourier) space $S(k_y)$. The profile $s(y)$ is obtained from the Fourier transform of $S(k_y)$. A frequency-encoded spin echo profile is obtained by applying a magnetic field gradient of amplitude G_y , such that excited nuclear spins precess with a frequency determined by their position. If the magnetic field gradient is applied for a time t , then the range of spatial wavenumbers $2k_{\text{max}}$ spanned in k -space is given by

$$\pm k_{\text{max}} = \pm \frac{\gamma G_y t}{2\pi}. \quad (7)$$

Each point in k -space will have a width defined by

$$\Delta k_y = \frac{\gamma G_y \Delta t}{2\pi}, \quad (8)$$

where Δt is the dwell time. The field of view (FOV) of the profile is then determined by the reciprocal relation $1/\Delta k$ such that the spatial resolution is $\Delta y = \text{FOV}/p$, where p is the number of data points acquired across the spin echo.

The frequency-encoded $y - T_2$ map pulse sequence is shown in Fig. 4. A gradient pulse of amplitude G_y is applied for a time $t/2$ to encode the spatial position between the initial 90° and 180° rf pulses. During the acquisition of the echo intensity, the gradient pulse is reapplied for a duration t to decode the spatial position. The application of the rf and gradient pulses is repeated to form an echo train akin to the CPMG T_2 measurement. Experimentally, $p = 64$, $\Delta t = 10 \mu\text{s}$, and $t_E = 2 \text{ ms}$.

Processing of the data is divided into the following steps. First, each of the n echoes is Fourier transformed. As part of this process it is desirable to include a first-order phase shift. Any deconvolution of the zero-gradient echo shape can be applied at this point. We have included a Blackman filter to remove truncation artifacts. The echoes are stacked to form a 2D data set such that each spatial position has an associated CPMG decay. Each spatial position is then treated as an individual 1D CPMG data set and inverted to form a T_2 distribution as described in Section 3.1. The resultant 2D map contains the $y - T_2$ correlation.

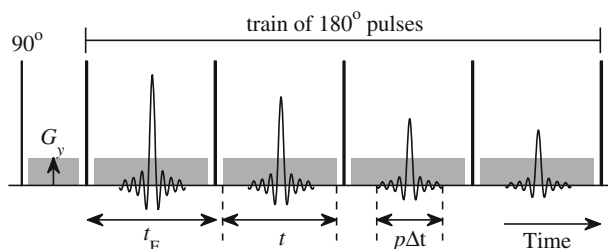


Fig. 4 Timing diagram for frequency-encoded $y - T_2$ mapping sequence. Vertical bars represent rf pulses. Gray rectangles represent magnetic field gradient pulses of constant amplitude G_y . The CPMG echo train shown is truncated for clarity

4 Results

To provide quantitative volumetric results from the NMR data, the bulk liquids (synthetic brine, crude oil, polymer solution) were measured with each of the acquisition protocols described in Sect. 3. These bulk liquid measurements allowed a calibrated relative signal intensity (RSI) per gram of sample to be determined at 46.4 °C. Each protocol provided a different RSI value for each liquid and the results are summarized in Table 1. The ASP solution had an RSI equal to the polymer solution. The 1D T_2 measurement appears to provide significantly more signal per unit gram than the 2D $T_1 - T_2$ or $D_{app} - T_2$ data. This difference is a consequence of the inversion algorithms applied to the 1D and 2D Fredholm integral equations, e.g., the incremental bin size in 1D $\log_{10} T_2$ is smaller than in the 2D $\log_{10} T_1$ versus $\log_{10} T_2$. This difference results in a modified scaling of the total amplitude in the distributions that is consistent across like-for-like experiments. Nevertheless, the differences in Table 1 highlight the importance of calibrating each measurement in order to determine an accurate saturation or porosity from NMR.

The NMR data were analyzed for fluid saturation by simple partitioning of (i) 1D T_2 relaxation data, (ii) 2D $T_1 - T_2$ relaxation data, and (iii) 2D $D_{app} - T_2$ diffusion–relaxation data. The oil/aqueous fluid-phase cut-off is illustrated in Figs. 5, 6, and 7 for the three protocols, respectively. Small variations in the cut-offs (being equivalent to one or two intervals in the chosen $\log_{10} T_2$, $\log_{10} T_1$, or $\log_{10} D_{app}$ scales) resulted in negligible changes in the estimated saturations.

Quantitative NMR saturations are obtained by integrating under the regions of the relaxation or diffusion distributions corresponding either to the oil or aqueous fluid-phases. The significance of this integration step is highlighted in Fig. 5: the T_2 distributions (top) obtained

Table 1 Relative signal intensity (RSI) of flood liquids at 46.4 °C

Protocol	RSI / g ⁻¹		
	Crude oil	Synthetic brine	Polymer solution
T_2	6047	4198	4593
$T_1 - T_2$	257	225	241
$D_{app} - T_2$	184	213	218

Differences are observed in the total signal obtained with each measurement protocol as a consequence of the differing sensitivities of each method and the details of the inversion algorithms

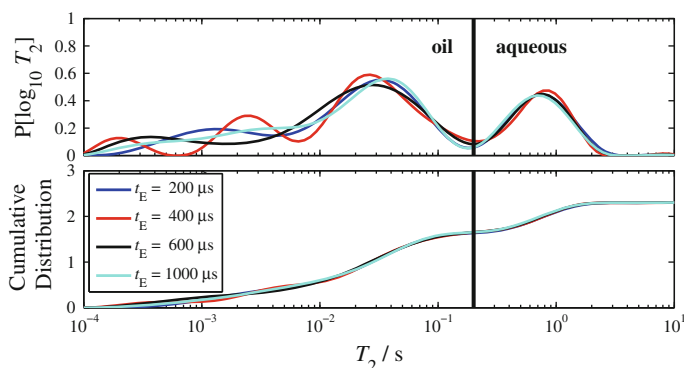


Fig. 5 Example analysis of 1D T_2 relaxation data showing (top) T_2 relaxation time distributions and (bottom) cumulative T_2 distributions. The legend applies to both plots. The solid vertical black line indicates the T_2 cut-off, chosen individually for each stage of the flood to obtain the optimum partitioning. A small overlap exists in the T_2 distributions of the oil and aqueous fluid-phases. Data were obtained after pumping 0.32 PV brine into plug B1

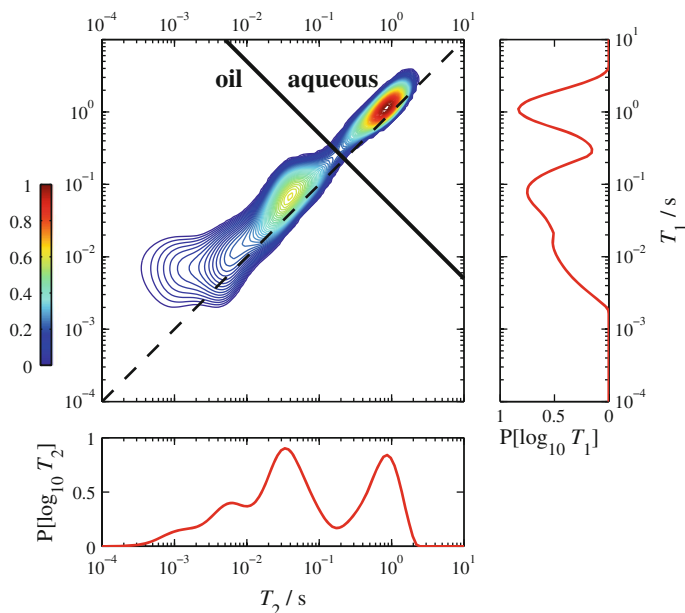


Fig. 6 Example analysis of 2D $T_1 - T_2$ relaxation data. The diagonal solid black line indicates the $T_1 - T_2$ cut-off, chosen individually for each stage of the flood. Some overlap between the oil and aqueous fluid-phase signals still exists in the $T_1 - T_2$ plane. The diagonal black dashed line indicates $T_1 = T_2$. The color scale is in arbitrary magnetization (NMR signal) units and is therefore proportional to fluid volume. Marginal (projected) plots are shown for clarity. Data were obtained after pumping 0.32 PV brine into plug B1. (Color figure online)

with different t_E echo spacings are qualitatively different, in particular for $T_2 < 10^{-2}$ s because the inversion process is sensitive to noise and weighted to small fluctuations in the initial echoes of the CPMG decay. For this reason, detailed interpretation of T_2 distributions can be misleading unless care is taken not to over-interpret the results. However, if we now consider the cumulative distributions in Fig. 5 (bottom) no significant differences are

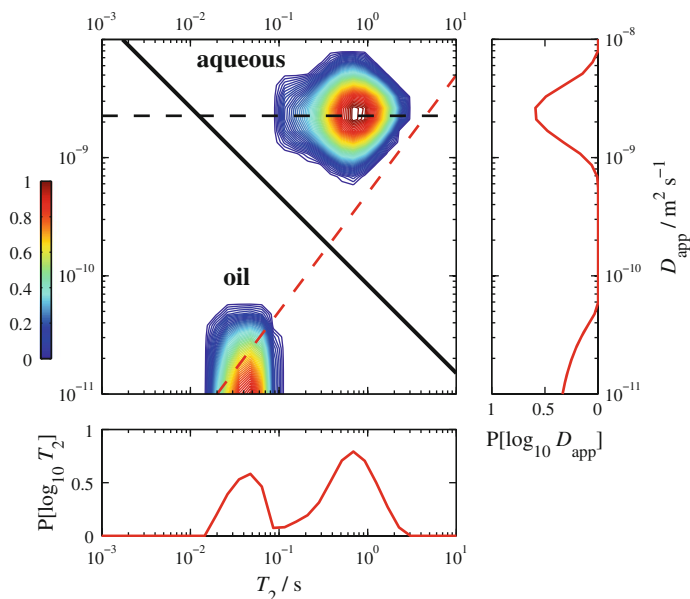


Fig. 7 Example analysis of 2D $D_{\text{app}} - T_2$ diffusion/relaxation data. The *diagonal solid black line* indicates the $D_{\text{app}} - T_2$ cut-off. The oil and aqueous fluid-phase signals are well separated in the $D_{\text{app}} - T_2$ plane. The *horizontal black dashed line* indicates $D_0 = 2.26 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$ for bulk water at 25 °C and the *diagonal red dashed line* is the canonical “alkane” diffusion–relaxation correlation. The *color scale* is in arbitrary magnetization (NMR signal) units and is therefore proportional to fluid volume. Marginal (projected) plots are shown for clarity. Data were obtained after pumping 0.32 PV brine into plug B1. (Color figure online)

observed between these data and consistent integral values (proportional to liquid volume) are obtained both at the $T_2 = 0.2 \text{ s}$ cut-off and for $T_2 > 2 \text{ s}$. The same concept applies to volume integration under the peaks in the 2D $T_1 - T_2$ and $D_{\text{app}} - T_2$ distributions.

In the bulk liquids (not shown), the oil and brine relaxation times are well separated. However, the inclusion of some light organic fractions with long T_1 and T_2 in the oil result in a small yet significant overlap between the oil and brine relaxation time distributions when in the rock; see Fig. 5 and 6. This overlap is due to the convolution of the effects of chemistry, mobility, and pore size distribution on the observed relaxation times. The overlap is particularly significant in the 1D T_2 distributions. The inclusion of a second dimension in the $T_1 - T_2$ correlations (Fig. 6) allows a more precise cut-off to be defined diagonally in the $T_1 - T_2$ plane, i.e., the minimum signal intensity in the 2D plot is less than that observed in either of the 1D T_1 or T_2 marginal distributions. The positioning of the cut-offs shown in Figs. 5 and 6 result in a slight over-estimate of the brine volume from the 1D T_2 distribution whereas the $T_1 - T_2$ correlation provides an under-estimate of the brine volume. In both cases, the oil saturation is found to be more accurate.

The 2D $T_1 - T_2$ correlation also provides the additional parameter of the T_1 / T_2 ratio. The ratio T_1 / T_2 is considered a qualitative probe of surface interaction strength (McDonald et al. 2005) which is independent of geometric parameters, unlike individual T_1 and T_2 measurements. The ability to measure T_1 / T_2 can be useful in determining the wettability of the rock (Mitchell et al. 2009). Here, the ratio for both the oil and aqueous fluid-phases was seen to be invariant (within the accuracy of the determination of T_1 / T_2 from the 2D plots) throughout

the oil recovery experiment. Accordingly, we draw no conclusions about variations, if any, in the wettability of the rock surface.

A very clear separation between the NMR signals associated with oil and brine is achieved in the $D_{\text{app}} - T_2$ correlation, see Fig. 7. The brine is seen to diffuse with its bulk diffusion coefficient. In highly tortuous porous media, the diffusion coefficients of confined liquids can be reduced by the physical restrictions imposed by the pore network. However, in the Bentheimer sandstone used in this study, the tortuosity is low and the pores large, so the diffusion of the brine is not restricted. The crude oil studied here, with a viscosity $\mu_o \sim 100$ mPa s, is on the lower limit of measurable diffusivity given its short T_2 relaxation time and the maximum pulsed gradient strength available on the NMR system. In general, short t_{SE} times will be required in the measurement of heavy oils, which are both fast-relaxing and slow-diffusing. Shorter t_{SE} times may be employed on a different magnet system with better eddy-current performance, and in particular on logging tools where a gradient stabilization time is not required at all. However, long t_{SE} times will still be required for heavy oils on all systems to achieve some diffusion weighting with the typical gradient strengths available.

The quality of the $D_{\text{app}} - T_2$ data, Fig. 7, is reduced due to the long gradient stabilization time (5 ms) required here to allow the eddy currents to stabilize following the gradient pulse. Eddy currents are ubiquitous in PFG experiments, particularly when using strong gradients. As such, reasonably long t_{SE} times are required to achieve reliable results. Combined with the short T_2 of the crude oil, these long echo times lead to a loss of signal. Notwithstanding, the oil volume can be determined because the calibrated RSI accounts for most of this lost signal. The only significant error in the oil saturation will occur if the lighter organic fractions (long T_2) are removed while the heaviest organic fractions (short T_2) remain. However, as the majority of the oil signal in the T_2 distribution lies between $T_2 = 10^{-2}$ s and 10^{-1} s, which can be probed thoroughly by this $D_{\text{app}} - T_2$ measurement, the errors are expected to be small and this assumption is borne out in the recovery data, see Sect. 4.1.

4.1 Oil Recovery Monitoring: NMR Versus Gravimetric

The three different NMR methods (namely T_2 , $T_1 - T_2$, $D_{\text{app}} - T_2$) gave slightly different calibrated results for absolute liquid volumes. Here, we consider the consolidated data. The liquid volumes observed in plugs B1 and B2 are shown in Fig. 8a and b, respectively. The reference pore volumes determined by He porosimetry are shown for comparison. The He pore volume is expected to be more accurate than the NMR total volume. All three NMR methods are based on spin echoes and are therefore expected to provide consistent results; the error bars shown are derived from the variation in calculated volumes between the NMR techniques. Notably, the brine volumes varied more between the data sets than the oil volumes. The error on the NMR data increased as the flood progressed due to the increased volume of electrically conductive liquid in the core-holder, in particular the ASP solution as noted in Sect. 2.1.

The total signal deficit in Fig. 8 is interesting primarily due to the near identical behavior observed in both floods. The initial total liquid volume is low by ~ 0.5 cm³ compared to the He pore volume. Due to the orientation of the loading procedure (placing the plug in the vertical core-holder), the plug is surrounded by air when confining pressure is applied. It is possible then that air is forced into the sides of the plug, displacing liquid at the outlet. As the brine flood progresses (0–0.64 PV), the total signal deficit increases in both experiments to ~ 1 cm³. Although the brine was outgassed prior to injection, some residual dissolved gases could have escaped due to the pressure drop across the plug. On commencement of the polymer drive (0.64–1.14 PV), the signal deficit is reduced. Some gas release was noted at

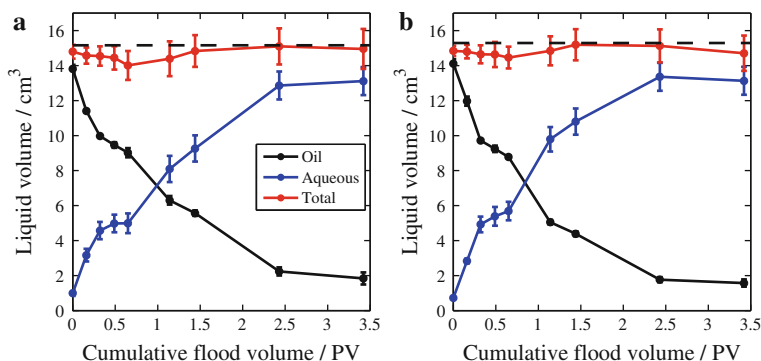


Fig. 8 Consolidated NMR estimates of liquid volumes in **a** plug B1 and **b** plug B2. The liquid volumes are determined from the NMR signal intensity, using a scaling factor obtained from a known volume of liquid. Each datum shown here is the average liquid volume measured by three different NMR methods: T_2 , $T_1 - T_2$, and $D_{app} - T_2$. The error bars indicate the variation in liquid volume obtained from the three different NMR methods; this variation arises from a number of factors, including the method of data processing and the RSI (see Table 1) associated with each technique. Also shown is total pore volume from He porosimetry (dashed horizontal line). Legend applies to both plots. Flood stages were brine (0–0.64 PV), polymer solution (0.64–1.14 PV), ASP solution (1.14–1.44 PV), and polymer solution (1.44–3.44 PV)

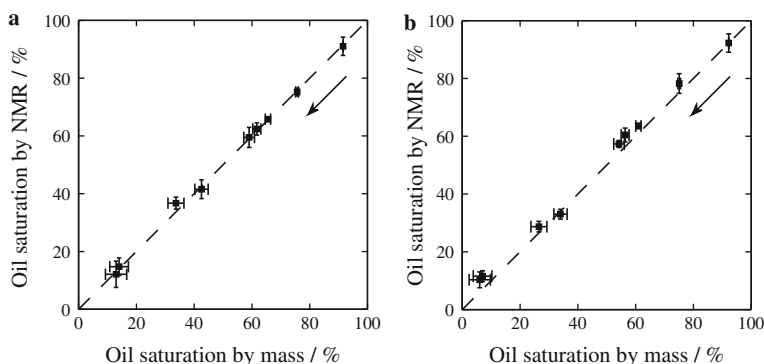


Fig. 9 NMR versus gravimetric determinations of oil saturation in **a** plug B1 and **b** plug B2. In each plot, the dashed diagonal is the identity line and the arrow indicates progress of the flood. Excellent agreement is observed between the two independent (NMR vs. gravimetric) estimates of remaining oil

the exit port during the polymer solution injection stages; the higher viscosity of the polymer solution may have been sufficient to displace trapped air or volatile oil components present as a gaseous fluid-phase.

NMR oil saturations are correlated against the gravimetric data in Fig. 9a and b for plugs B1 and B2, respectively. The consolidated NMR results from Fig. 8 were used to generate these plots. The general agreement is excellent throughout both experiments, and the two sets of measurements provide a 1 : 1 correlation within the estimated errors of the remaining oil saturations. For the gravimetric results, an error of 50 mg was assumed at each stage. These errors are cumulative as the saturation is determined from the sum of the recovered oil up to that point. Although the error on any one of the NMR saturation values is larger than the equivalent of 50 mg oil, each NMR measurement is an independent determination of oil saturation; therefore the errors do not accumulate. Hence the NMR analysis has the

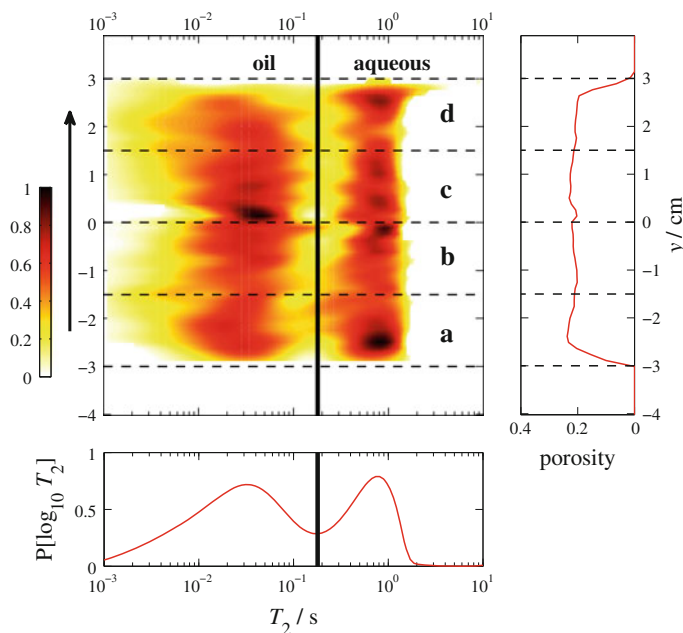


Fig. 10 Example analysis of $y - T_2$ spatially resolved relaxation data. The low resolution and SNR observed in the profile dimension is a consequence of acquiring the frequency-encoded profile using a low field NMR spectrometer. Variations in the total signal intensity (porosity) result predominantly from non-ideal gradient performance rather than heterogeneities in the rock. The *solid vertical black line* indicates the T_2 cut-off, chosen individually for each stage of the flood. The *dashed horizontal lines* indicate the division of the y -axis into four equal quarters labeled *a* (inlet) to *d* (outlet). The *vertical arrow* indicates the direction of flow. The *color scale* is in arbitrary magnetization (NMR signal) units and is therefore proportional to fluid volume. Marginal (projected) plots are shown for clarity. Data obtained after pumping 0.32 PV brine into plug B1. (Color figure online)

potential to provide a more accurate determination of final oil saturation than the conventional gravimetric analysis.

4.2 Spatially Resolved Oil Saturation

Based on piloting results published elsewhere for a similar combination of EOR chemicals (Arora et al. 2010), we would expect $S_o^{(final)} \rightarrow 0\%$. However, the average final oil saturation observed in both B1 and B2 was $S_o^{(final)} \approx 10\%$. To explain this discrepancy, spatially resolved T_2 distributions were examined. A typical $y - T_2$ map is shown in Fig. 10. Separation between the oil and aqueous fluid-phase signals was achieved from the stacked T_2 distribution (marginal plot) by placing a cut-off at $T_2 \approx 0.2$ s. The y profile (marginal plot) shows the total liquid signal (porosity) and the profile remained rectangular throughout the flood, as expected.

The oil saturation was determined over four quarters of the y -axis, as indicated by a–d in Fig. 10. These integral limits were chosen to provide spatial resolution when determining localized S_o values without the results being dominated by noise fluctuations. The resolution and SNR achievable in our experiments are much lower than reported commonly in the literature for spatially resolved MRI measurements of core-plugs. However, we remind the reader that, while spatial resolution on the order of millimeters can be obtained on typical

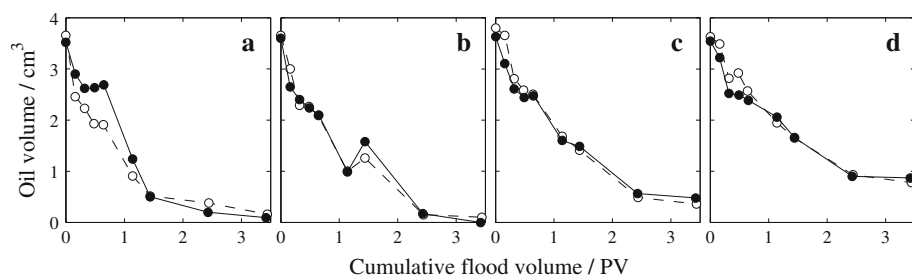


Fig. 11 Spatially resolved oil fluid-phase saturations in experiments B1 (*solid symbols and lines*) and B2 (*open symbols and dashed lines*) shown for each quarter of the plug from the (a) inlet to (d) outlet. The panels correspond to the spatial quarters *a* to *d* shown in Fig. 10. An increase in oil saturation is observed clearly around 1 PV cumulative flood volume in *a* and *b* as oil is redistributed by the injection front. This process of resaturation with oil is referred to as the transport of an oil bank through the plug

MRI magnets (as employed in medical diagnosis), we are using a very low field instrument in this study. Obtaining spatial information on a 2 MHz “rock core analyzer” is atypical (based on the dearth of equivalent experiments in the literature) and the quality of the profiles reflect the technical difficulties associated with such acquisitions. Notwithstanding, we are able to observe the significant variations in oil saturation as a function of position in these core-plugs. The variation in oil saturation through the course of the recovery experiments in each of the spatial quarters are shown in Fig. 11. It is obvious from these spatially resolved data that the oil recovery did not occur uniformly along the length of the core-plug. Near the inlet, Fig. 11(a, b), almost complete oil recovery ($S_o^{(final)} \rightarrow 0\%$) is observed by the end of the flood. However, significant remaining oil is seen at the outlet, Fig. 11d, where $S_o^{(final)} \approx 20\%$.

In EOR studies of short core-plugs, capillarity can play a major role in distribution of remaining oil. Capillary trapping is expected to occur in microscopically heterogeneous, low permeability pore networks (Yortsos and Chang 1990; Yortsos et al. 2001; van Duijn et al. 2002). Capillary end effects arise due to differences in permeability and wettability between the core-plug faces and the distribution plates of the core-holder, according to the theory by van Duijn et al. [1995]. However, the Bentheimer rock used in this study has a very high permeability, large pores, and is preferentially water wet. Therefore, we associate the non-uniform remaining oil saturation with viscous instabilities, enhanced by the high linear flow rates and core-holder geometry. Viscous instabilities have been observed previously in this formation during conventional laboratory studies (Bouwmeester and Faber 2012). In future, we will monitor oil recovery processes at lower injection rates to determine the influence of flow velocity on the spatial distribution of remaining oil.

Additional information about the oil recovery process can be obtained from the spatially resolved measurements. For example, in Fig. 11 the progress of an oil bank is observed as an increase in S_o between 0–1 PV flood volume in (a) and 1–1.5 PV in (b). No strong evidence of an oil bank is observed in the latter half of the plug, Fig. 11 (c, d), and this may be a consequence of the altered flow behavior. Overall, we consider Fig. 11(b) to provide the most accurate representation of oil recovery from an “infinite” volume (i.e., an oil reservoir) as the influence of entry and exit geometry appears minimal over this quarter of the plug.

An Eclipse³ simulation of the EOR process was performed to validate the viscous fingering observed in the experiment. This simulation uses the standard continuum model for

³ Eclipse family of reservoir simulator components is a Mark of Schlumberger

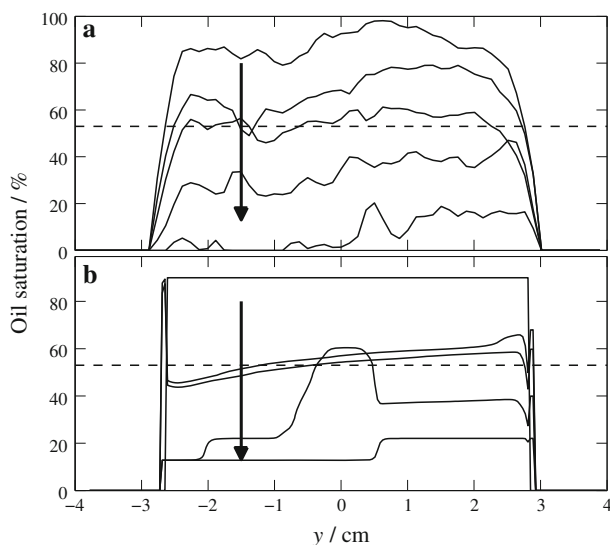
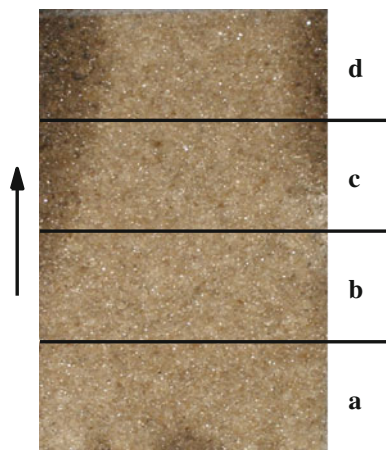


Fig. 12 Remaining oil saturation profiles obtained at various flood stages from **a** NMR $y - T_2$ maps and **b** eclipse reservoir simulation. Profiles correspond to total injected fluid volumes of (top to bottom) 0, 0.16, 0.65, 1.14, and 3.42 PV. The arrows indicate the progression of the flood. The average oil saturation at end of the brine injection/start of the polymer solution injection is indicated by the horizontal dashed line in each plot

immiscible multi-phase fluid flow in porous media, based on the established concepts of relative permeability and capillary pressure (Marle 1981). In this study, we perform a 2D simulation in a horizontal plane (no gravitational effects). The simulation “reservoir” domain was designed to include the geometry of the distribution plates on the core-holder at the inlet (injection) and outlet (production) faces of the plug. The permeability of the rock was set at $3.3 \mu\text{m}^2$ and the permeability of the flow channels in the oil wet distribution plates was $10 \mu\text{m}^2$. Generic relative permeability functions were used, with the relative permeability end points adjusted to match the experimental results for final oil saturation. The rock was assumed to be water wet with $S_o^{(\text{initial})} = 90\%$. MIP data were rescaled using the known IFT values to obtain estimates of the capillary pressure curves. The viscosity of the injection “water” was set to either 1 mPa s or 100 mPa s to simulate the brine and polymer solution injection stages of the flood, respectively. Linear flow rates were selected to match the experimental conditions. This Eclipse simulation did not take into account the presence of the surfactant. A selection of simulated profiles is compared to the experimental results obtained from plug B1 in Fig. 12. The final oil saturation was higher in the simulation than the experiment likely due to the absence of the surfactant. Nevertheless, the salient features of the experimental profiles are reproduced in the simulation. The brine fingers immediately through the heavy oil, leaving an almost uniform distribution of remaining oil at the end of the brine flood stage. No significant capillary end effect occurred on brine injection. The high viscosity polymer solution is seen to displace much of the remaining oil. In the simulation, an oil bank is observed on polymer injection, supporting the resaturation of the plug observed experimentally in Fig. 11b; unfortunately, this feature is not well resolved in the corresponding NMR profile in Fig. 12a. The polymer solution follows the flow path created by the brine, displacing additional oil in the process, resulting in geometry-dependent end effects in the final oil saturation. The differences between simulation and experiment notwithstanding,

Fig. 13 Photograph of plug B1 cross-section at the end of the flood. Remaining oil (*dark brown*) is seen in contrast to the clean rock (*light brown*). The plug quarters labeled *a* (inlet) to *d* (outlet) correspond to the same divisions in Fig. 10 and plots (*a*) to (*d*) in Fig. 11. The *vertical arrow* indicates the direction of flow. (Color figure online)



the Eclipse result supports the main conclusions drawn from the spatially resolved NMR data. Oil remains trapped at the plug outlet, and to a lesser extent at the inlet, due to viscous instabilities that occur when the low viscosity water is injected into the high viscosity oil. The introduction of high viscosity polymer improves the oil recovery, but the geometry of the core-holder prevents $S_o^{(\text{final})} \rightarrow 0$ throughout the entire length of the plug.

To provide visual confirmation of the spatially resolved oil saturations observed in the NMR data and simulation, plug B1 was split in half and the cross-section examined, see Fig. 13. The remaining oil is seen clearly as a dark coloration of the sandstone. Assuming radial symmetry (based on visual inspection of the plug end faces), oil saturation estimations were made from the image in Fig. 13. A simple threshold was imposed to attribute an oil saturation to each pixel: $S_o^{(\text{final})} = 0\%$ (light brown) or $S_o^{(\text{final})} = 40\%$ (dark brown). The number of “clean” and “oily” pixels in quarters a to d were multiplied by an appropriate cylindrical function to provide a volume and the average saturation in each quarter calculated. The remaining oil saturations were (a) $S_o^{(\text{final})} = 1.5\%$, (b) $S_o^{(\text{final})} = 0.7\%$, (c) $S_o^{(\text{final})} = 12\%$, and (d) $S_o^{(\text{final})} = 22\%$. These values are in good agreement with the NMR final oil saturations for each quarter shown in Fig. 11, supporting the conclusion that the non-invasive, spatially resolved NMR data represent the remaining oil distribution accurately.

5 Conclusions

The quantitative nature of low field NMR relaxation and diffusion measurements has been demonstrated using the example of an in situ EOR study. Excellent agreement was observed between NMR and conventional gravimetric estimates of remaining oil. The NMR measurement provides a direct volumetric determination of both the oil and aqueous fluid-phases (brine, polymer solution, and ASP solution) present in the core-plug, whereas only the oil saturation could be estimated based on gravimetric assays of the effluent. For the heavy oil studied here, good fluid-phase discrimination was achieved based on “diffusion-editing” $D_{\text{app}} - T_2$ correlations using a NMR logging-tool equivalent pulse sequence. Sufficient fluid-phase discrimination was also achieved in the relaxation measurements (1D T_2 distributions and 2D $T_1 - T_2$ correlations) to provide oil saturations comparable to the gravimetric results. The NMR measurements, being sensitive to the total liquid volume, indicated a signal deficit

relative to the known plug volume that was attributed to air or volatile oil vapor trapped in the plug. This information on total liquid saturation cannot be obtained readily with gravimetric analysis.

The bulk NMR measurements revealed an average final oil saturation of $S_o^{(final)} \approx 10\%$. However, it was expected from a related pilot operation (Arora et al. 2010) that $S_o^{(final)} \rightarrow 0\%$ should be observed. Spatially resolved $y - T_2$ maps showed that near-complete oil recovery was achieved in the first half (inlet) of the plug. However, preferential flow behavior at the outlet meant the remaining oil saturation increased toward the exit face, accounting for the unexpectedly high bulk $S_o^{(final)}$. It is likely that the geometric effects appear more significant due to the small plug length. Therefore, these results highlight the importance of spatial resolution in any oil recovery monitoring on such small plugs. In this regard, NMR offers the advantage of spatial resolution that can be combined with other data acquisitions such as relaxation time distribution measurements.

In future work, we will study oil recovery achieved at lower interstitial flow rates of the aqueous fluid-phase, relevant to oil reservoirs. A comparison between our experimental results and computational simulations of multi-phase flow-through porous media will provide insights into the effect of geometry on the oil recovery process. These simulations may allow for modifications to the core-holder design, thereby minimizing the formation of preferential flow paths through the plug. These modifications will be of significance to any flow-through core analysis using liquids, regardless of the measurement techniques employed.

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